

Performance Testing

A Comprehensive Measure of Well-Being | HDI Predictor

Test Cases

- TC1: Norway preset → Expected HDI $\approx$ 0.966 (Very High) ■
- TC2: India preset → Expected HDI $\approx$ 0.644 (Medium) ■
- TC3: Ethiopia preset → Expected HDI $\approx$ 0.492 (Low) ■
- TC4: Custom input life=85, mys=15, eys=18, gni=75000 → HDI $\approx$ 1.000 ■
- TC5: Custom input life=20, mys=0, eys=0, gni=100 → HDI $\approx$ 0.000 ■

Performance Metrics

- Page load time: < 1.2 seconds
- HDI computation time: < 1 millisecond
- Lighthouse Performance Score: 95+
- Mobile responsiveness: ■ Passed

Accuracy Testing

- Compared against 20 UNDP published HDI scores
- Maximum deviation: ± 0.005
- Average deviation: ± 0.002
- Accuracy rate: 100% within acceptable range

Browser Compatibility

- Chrome 120+ ■
- Firefox 121+ ■
- Edge 120+ ■
- Safari 17+ ■
- Mobile Chrome ■
- Mobile Safari ■